# NOTE TEMPLATE EXAMPLE

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Abstract. This example demonstrates the basic structure of the note template: sections, theorem-like environments, cross-references, displayed equations, lists, commutative diagrams, citations, and a bibliography.

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## 1. A first section

This is ordinary mathematical text. We cite a standard reference [Har77] and refer to Theorem 1.2 below. Later we also use the lowercase form proposition 1.3.

**Definition 1.1.** A *sample object* is an object introduced only to demonstrate the formatting of definitions.

**Theorem 1.2.** *Let $X$ be a sample object. Then the following properties hold:*

*(i) $X$ has a first property.*
*(ii) $X$ has a second property.*

*Proof.* This is a sample proof. The displayed equation

$$(a+b)^2 = a^2 + 2ab + b^2$$

illustrates ordinary mathematical display style. □

By Item (i), the first property holds.

**Proposition 1.3.** *The sample object of Definition 1.1 also satisfies a proposition.*

**Lemma 1.4.** *This is a sample lemma.*

**Corollary 1.5.** *This is a sample corollary.*

*Remark* 1.6. This is a remark. It demonstrates the remark style.

**Example 1.7.** This is an example environment.

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## 2. A diagram

The template currently supports commutative diagrams using `tikz-cd`:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D. \end{array}$$

## 3. References

We have used Definition 1.1, theorem 1.2, and proposition 1.3 and cited [Har77].

## References

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer, 1977.